



PRATUL SHROFF
CEO AND FOUNDER, eINFOCHIPS

“The biggest design challenge for the Internet of Things is to keep up with the moving specifications”

The Internet of Things (IoT) is based on the idea of ambient intelligence, where hundreds of sensors pulling in data help a central system to make informed decisions and automate the functioning of a variety of systems. You will find it in your smart grid, intelligent farming, home/office automation and even on Google Maps!

Pratul Shroff, CEO and founder, eInfoChips, spoke to Dilin Anand and Abhishek Mutha of EFY about micro-robotics and the IoT connection, IoT product design challenges and solutions

What exciting areas do you see with the IoT concept?

One area of my immense interest is micro-robotics. We all have seen a dragon fly, right? Well, today micro-robots have been designed to the size of dragon flies, which can be unleashed in huge numbers for data gathering. There are also water-borne micro-robots that can perform similar function under water. This has huge implications with respect to the newer processors, sensors, integrated wireless connectivity and other features. These devices are capable of surveillance and data collection in huge amounts while maintaining a natural environment as they are too small to disrupt the environment.

How has IoT refreshed the concept of home automation?

Home automation system has a central device connected

to many sensors that monitor everything from hazardous gases, smoke, fire and flood to other potentially harmful conditions. Additionally, they can provide precisely controlled climatic conditions at home as well as the ability to manipulate all of this remotely.

Wireless integration, which is available now, has also allowed us to accept the data coming in from various machines in the house, such as the dishwasher, fridge, oven and TV. It can also accumulate your past data and use the historic information to advise you on potential power savings that can be made by changing the temperature, or keep you updated on the stock of some particulars in the fridge or mini-bar. Ultimately, it will make you aware of different expenditure and consumption patterns, enabling you to act upon them.